Universal gravitation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict how gravitational force changes with mass and separation.

SUCCESS CRITERIA - I can...

1. Use center-to-center separation.
2. Apply direct mass and inverse-square distance comparisons.
3. Identify equal forces on the two interacting bodies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gravity” and “separation” to explain your model or result. Sentence starters are optional.

gravity: an attractive interaction between masses

separation: distance between the specified objects or centers

inverse square: proportional to one divided by distance squared

01 Warm-up: bring back an earlier idea

A 3 kg object moves at 5 m/s in an ideal circle under a force perpendicular to velocity.
Explain which parts of velocity change and whether the force does work.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F = Gm_1m_2/r^2$.
r is center-to-center separation. At fixed masses, doubling r makes the attractive force one quarter as large.

Worked reasoning: If separation doubles while both masses stay fixed, gravitational force becomes $1/2^2=1/4$ as large. A 20 N interaction would become 5 N.

Your turn: Sketch the two separations using centers.
Draw equal and opposite force arrows for each pair.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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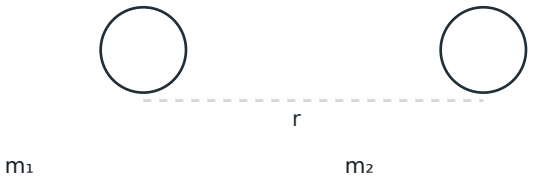
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two equal spheres are shown.
Mark the distance r used by the gravity equation and draw a force on each sphere.

Sentence starter: The … in my model represents … . Its label shows … .



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: 1 2 3

04 Apply the physics

A gravity simulator begins with attraction F_0. Multiply the first mass by 3, the second by 2, and the center-to-center distance by 3. Complete the multiplier model and find F/F_0.

Sentence starter: I found … [unit]. This value means … in this situation.

Factor in $F = Gm_1m_2/r^2$	Multiplier
m_1	3
m_2	2
r^2	___

Marzano check - circle: 1 2 3

05 Explain your reasoning

Does a large planet exert more force on a small moon than the moon exerts on the planet?
Compare forces and accelerations.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A spacecraft of mass 4 kg pushes an asteroid of mass 400 kg with 24 N.
Compare interaction forces and the accelerations due to that interaction.
Put each partner force on the object it acts on.

Word help: interaction = two objects affecting one another

Sentence starter: Both … . They differ in … because … .



Object A



Object B

Marzano check - circle: ① ② ③

07 Exit check

What separation makes force 1/9 of its original value? What if both masses also triple?

Sentence starter: I notice … . This supports … because … .



Mark the new separation.

Mass 1 Mass 2

Marzano check - circle: ① ② ③

08 STRETCH: test the model

Both masses triple. What separation multiplier keeps the force unchanged? What multiplier makes it half the original force?

Sentence starter: I notice … . This supports … because … .

Quantity	Original	New
m_1	m_1	____
m_2	m_2	____
r	r	____
Force	F_0	____

Marzano check - circle: ① ② ③